

Monitoring signal	**Estimates**	**Separates e from c?**
AI vs. final decision	$\mathbb{E}[e\,c + (1-e)\,o]$	no; looks better as c falls
Concordance (override rate)	$1 - \mathbb{E}[e\,c + (1-e)\,o]$	no
Delayed reference truth	$\mathbb{E}[e]$ on positives, lagged	e only
AI-blinded audit double read	$\mathbb{E}[e]$, and c on audited errors	yes, but costly
Projected sentinel cases	c on known AI errors	c only, efficiently

Fig. 1. Error-interception structure and what each monitoring signal observes.

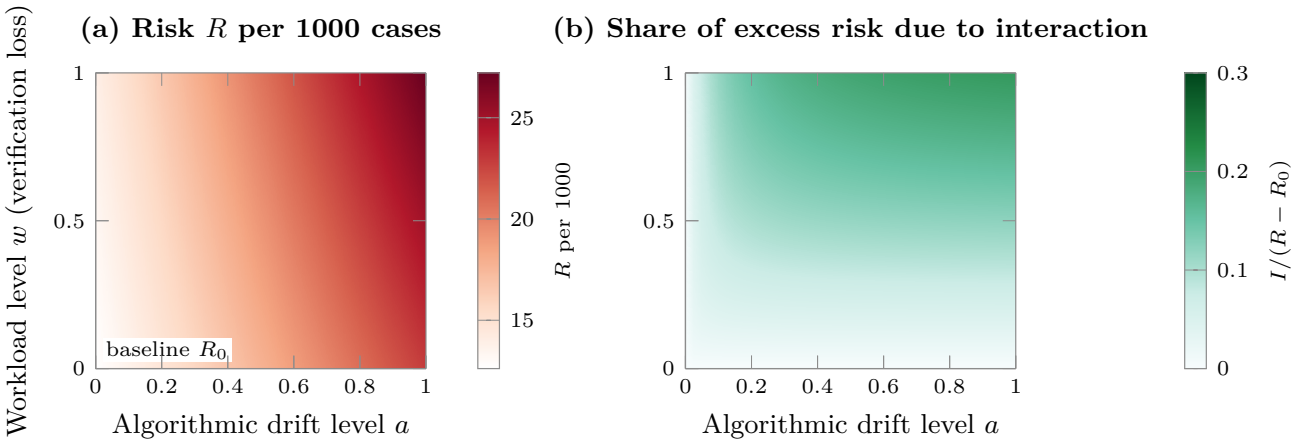


Fig. 2. Exact risk over drift levels and share due to interaction (E1).

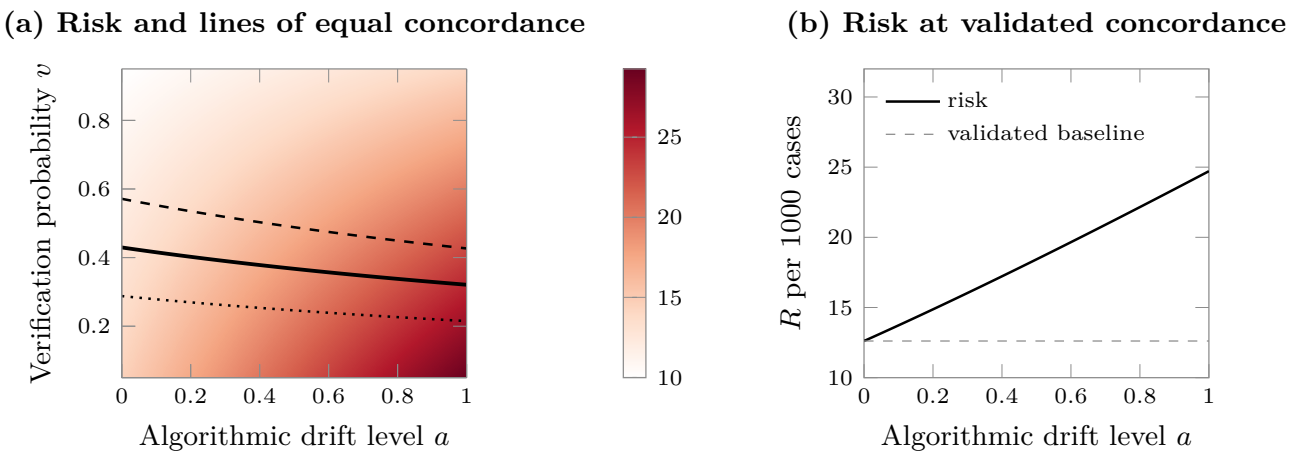


Fig. 3. Concordance does not identify risk (E2).

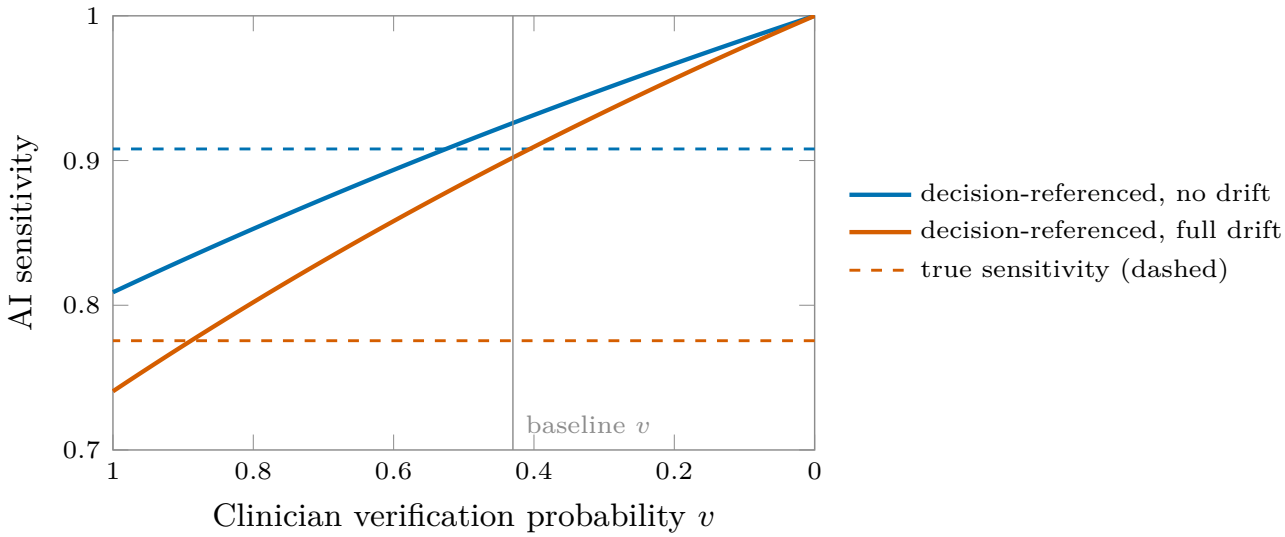


Fig. 4. Incorporation bias of report-referenced AI sensitivity (E3).

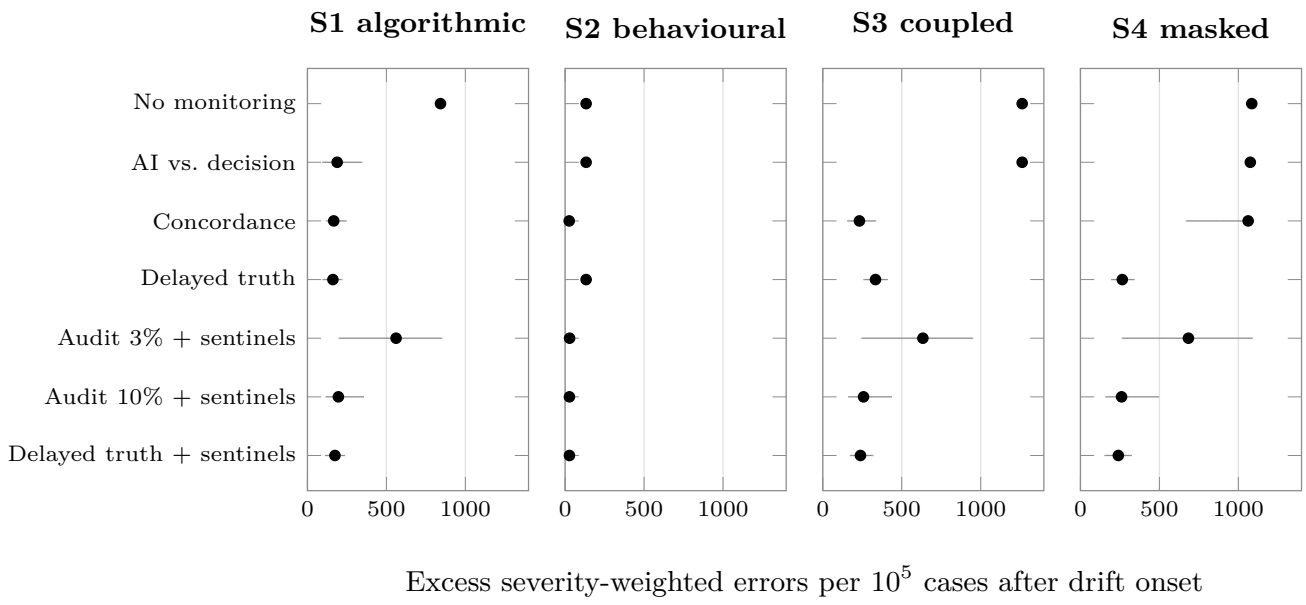


Fig. 5. Excess harm after drift onset by scenario and strategy (E4).

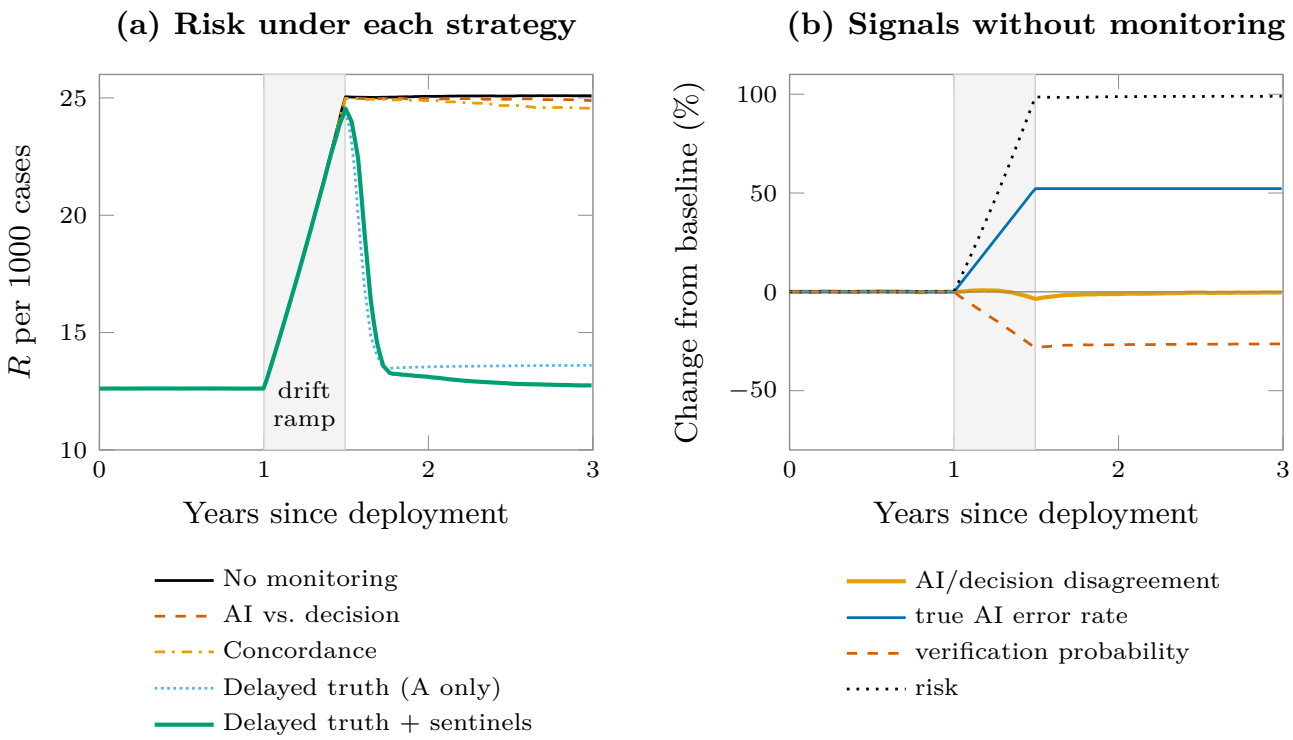


Fig. 6. Masked coupled drift: risk by strategy and monitored signals (E4, S4).

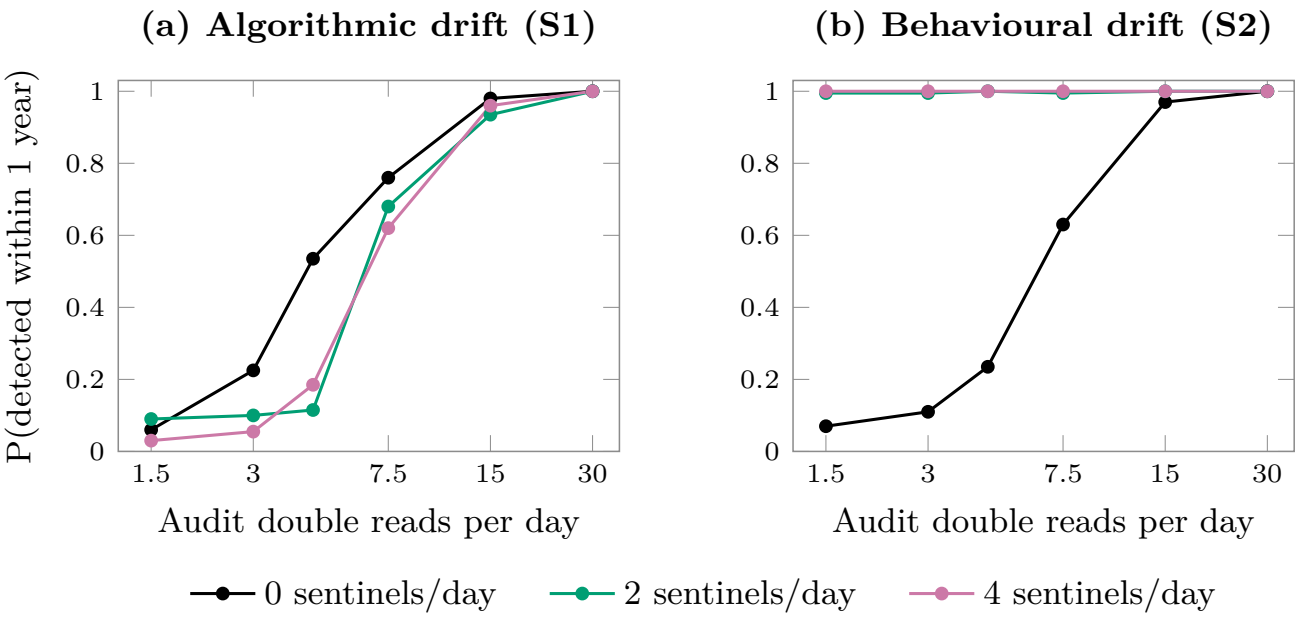
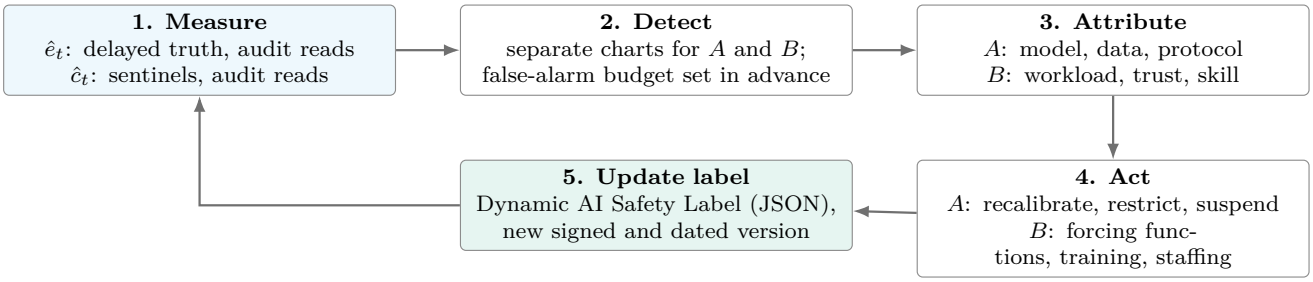


Fig. 7. Audit and sentinel design: detection within one year (E5).



Data sources in the imaging department

AI results	DICOM Structured Report; IHE AI Results (AIR) profile
Final decision	report in RIS; HL7 FHIR DiagnosticReport
Reader actions	viewer telemetry (open, accept, edit, time); FHIR AuditEvent
Reference truth	pathology, cancer registry, follow-up imaging
Sentinels	QA worklist cases with projected AI errors (synthetic or retrospective)

Fig. 8. Algorithmovigilance cycle and imaging informatics data sources.

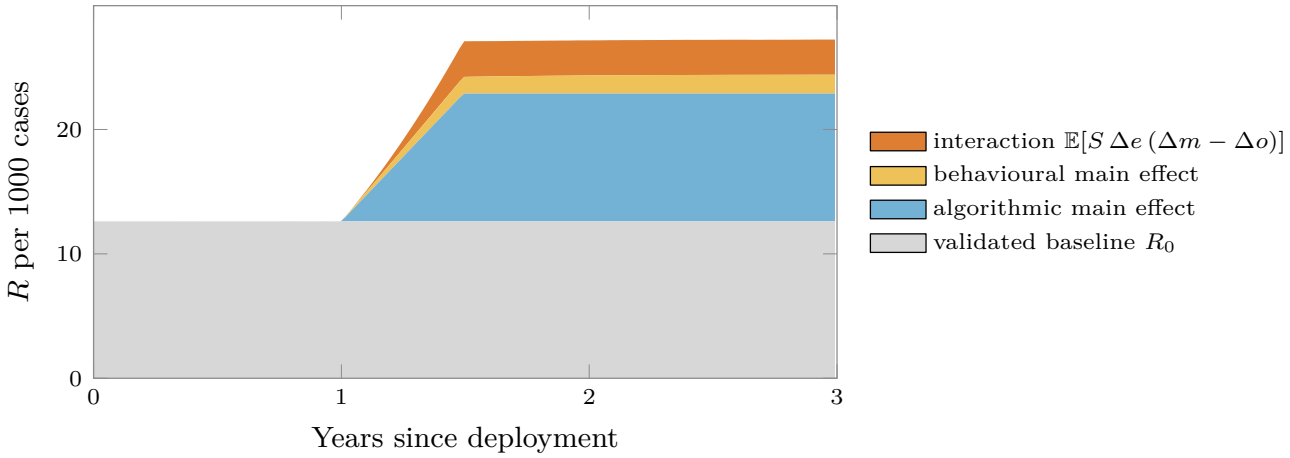


Fig. S1. Exact risk decomposition over time, coupled drift, no monitoring (E4, S3).